

Midterm 1 Practice

1. Use truth table to determine whether the following two statements are logically equivalent: $(P \Rightarrow Q) \wedge (P \Rightarrow R)$ and $P \Rightarrow (Q \wedge R)$.

2. Negate the following: $\neg Q \rightarrow P$
(*Show proper steps*) .

3. Which of the following matrices are in RREF? **4 points**

(Explain why/why not)

$$\begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

4. Use Gauss Jordan Elimination to solve the following system of equations.
(Show proper steps)

$$x_1 + 5x_3 = -4$$

$$x_1 + 4x_2 + 3x_3 = -2$$

$$2x_1 + 7x_2 + x_3 = -2$$

5. The sum of two numbers is 25. Their difference is 7. Find the numbers.

Solve using Gauss Jordan elimination and show proper steps

6. Prove using mathematical induction that for all $n \geq 1$: **10 points**

$$2 + 5 + 8 + \dots + (3n - 1) = \frac{n(3n+1)}{2}$$

(Write each step clearly)

7. Determine if the following vectors are orthogonal:

$$\vec{a} = (2, -1, 0)$$

$$\vec{b} = (1, 2, 3)$$

8. Multiply the following matrices:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 \\ 1 & 4 \\ 3 & -2 \end{bmatrix}$$

9. Let P be “A triangle is equilateral” and Q be “ A triangle is isosceles”

- Is P necessary or sufficient for Q?
- Write your answer as an implies statement

10. Given: "For every integer x , if x is *even* then x^2 is *even*".

- Write the above statement using mathematical notations.
- Write the negation of the above statement